

Git and GitHub -02

1. Install Git

```
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/techie-05/techie-05/.git (main)
$ git --version
git version 2.37.1.windows.1

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/techie-05/techie-05/.git (main)
$
```

2. Create a repo in github with README.md and .ignore file.

 .ignore file	Create .ignore file	1 n
 README.md	Create README.md	

3. Clone the created repo to local.

```
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/techie-05 (main)
$ git clone -b main git@github.com:Arunsai873/techie-05.git
Cloning into 'techie-05'...
remote: Enumerating objects: 6, done.
remote: Counting objects: 100% (6/6), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (6/6), done.

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/techie-05 (main)
$ cd techie-05

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/techie-05/techie-05 (main)
$ ls
test
```

4. Create two files in local repo.

```
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/techie-05/techie-05/.git (main)
$ touch a.txt b.txt
```

5. Commit two files and push to central Repository.

```
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git add .

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git commit -m "two files added"
[master 1ae2f1d] two files added
3 files changed, 1 insertion(+), 4 deletions(-)
create mode 100644 a.txt
create mode 100644 b.txt

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git push -u origin main
error: src refspec main does not match any
error: failed to push some refs to 'https://github.com/Arunsai873/'

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git push -u origin master
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
```

6. Create a branch in local and create a sample file and push to central.

```
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (release)
$ ls
hello

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (release)
$ git checkout master
Switched to branch 'master'

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ ls
hello
```

```
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ touch master

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git add .

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git commit -m "added file in master"
[master ed1ed72] added file in master
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 master

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ ls
hello master
```

```

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git push -u origin master
Enumerating objects: 29, done.
Counting objects: 100% (29/29), done.
Delta compression using up to 8 threads
Compressing objects: 100% (22/22), done.
Writing objects: 100% (29/29), 2.48 KiB | 508.00 KiB/s, done.
Total 29 (delta 6), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (6/6), done.
remote:
remote: Create a pull request for 'master' on GitHub by visiting:
remote:   https://github.com/Arunsai873/techie-05/pull/new/master
remote:
To https://github.com/Arunsai873/techie-05.git
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.

```

Active branches					
Branch	Updated	Check status	Behind	Ahead	Pull request
master	now		9	11	

7. Create a branch in github and clone that to local.

master had recent pushes 2 minutes ago

arun 4 Branches 0 Tags

This branch is 8 commits behind main .

Arnsai873 test

test test

Clone

HTTPS

git@github

```

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git clone -b arun git@github.com:Arnsai873/techie-05.git
Cloning into 'techie-05'...
remote: Enumerating objects: 59, done.
remote: Counting objects: 100% (59/59), done.
remote: Compressing objects: 100% (35/35), done.
remote: Total 59 (delta 13), reused 43 (delta 8), pack-reused 0
Receiving objects: 100% (59/59), 8.16 KiB | 759.00 KiB/s, done.
Resolving deltas: 100% (13/13), done.
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$

```

8. Merge the created branch with master in git local.

```

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git merge release
Auto-merging techie
CONFLICT (add/add): Merge conflict in techie
Automatic merge failed; fix conflicts and then commit the result.

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master|MERGING)
$ git status
On branch master
You have unmerged paths.
  (fix conflicts and run "git commit")
  (use "git merge --abort" to abort the merge)

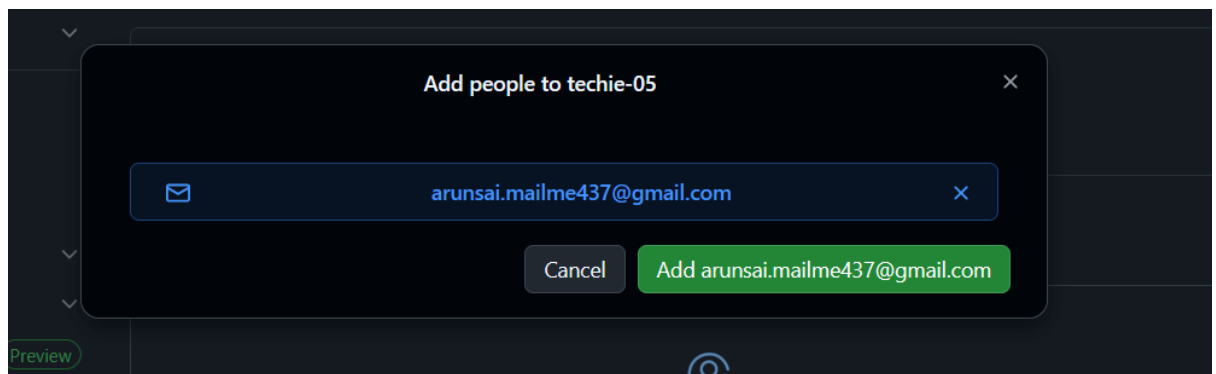
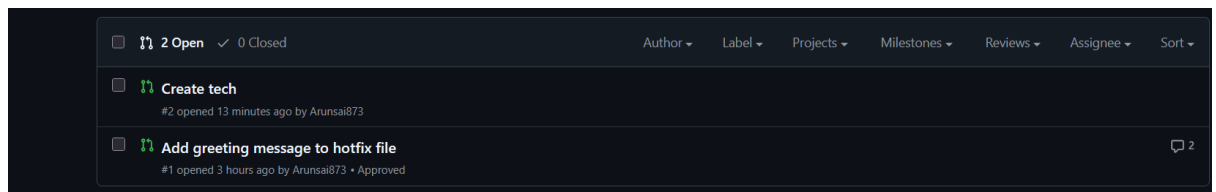
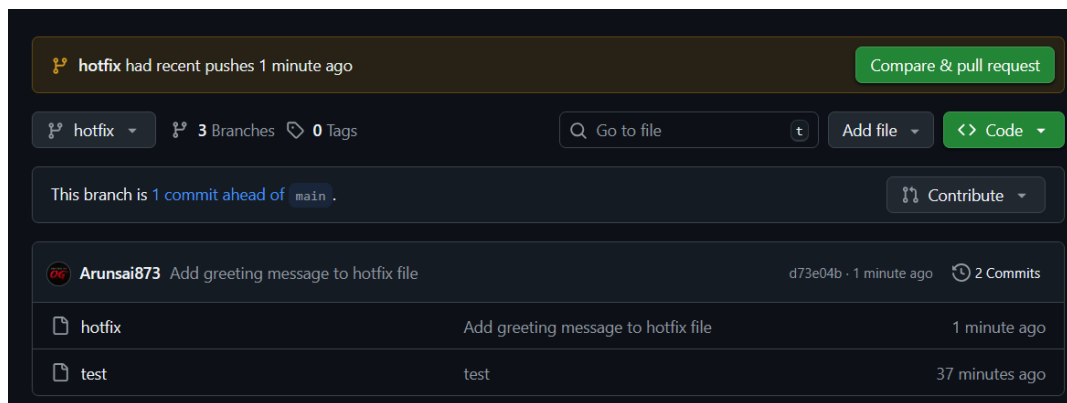
Changes to be committed:
  modified:   alphabet
  new file:   release

Unmerged paths:
  (use "git add <file>..." to mark resolution)
  both added:   techie

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master|MERGING)
$

```

9. Merge the created branch with master in github by sending a pull request.



10. Create a file in local and send that to branch in github.

```
arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (release)
$ git checkout master
warning: unable to rmdir 'techie-05': Directory not empty
Switched to branch 'master'
Your branch is up to date with 'origin/master'.

arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ touch a

arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git add a

arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git commit -m "a added in master"
[master d6e7854] a added in master
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 a

arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git push -u origin master
Enumerating objects: 3, done
```

11. Clone only a branch from github to local

```
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app/techie-05 (arun)
$ git branch
* arun

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app/techie-05 (arun)
$ git clone -b sai git@github.com:Arunsai873/techie-05.git
Cloning into 'techie-05'...
remote: Enumerating objects: 74, done.
remote: Counting objects: 100% (74/74), done.
remote: Compressing objects: 100% (48/48), done.
remote: Total 74 (delta 20), reused 46 (delta 10), pack-reused 0 (from 0)
Receiving objects: 100% (74/74), 13.47 KiB | 1.22 MiB/s, done.
Resolving deltas: 100% (20/20), done.

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app/techie-05 (arun)
($ ls
techie-05/ test

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app/techie-05 (arun)
$ cd techie-05

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app/techie-05/techie-05 (sai)
$ git branch
* sai
```

12. Create a file with all passwords and make that untrackable with git.

```
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/techie-05/techie-05 (main)
$ ls
password test

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/techie-05/techie-05 (main)
$ vi .gitignore

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/techie-05/techie-05 (main)
$ ls -a
./ ../ .git/ .gitignore password test

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/techie-05/techie-05 (main)
$ cat .git
cat: .git: Is a directory

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/techie-05/techie-05 (main)
$ cat .gitignore
password
```

```

Arunasai@DESKTOP-N6MPCBS MINGW64 ~/techie-05/techie-05 (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .gitignore

nothing added to commit but untracked files present (use "git add" to track)

Arunasai@DESKTOP-N6MPCBS MINGW64 ~/techie-05/techie-05 (main)
$ git add .
warning: in the working copy of '.gitignore', LF will be replaced by CRLF the next time
you touches it

Arunasai@DESKTOP-N6MPCBS MINGW64 ~/techie-05/techie-05 (main)
$ git commit -m "added .gitignore"
[main 1182c9b] added .gitignore
1 file changed, 1 insertion(+)
create mode 100644 .gitignore

Arunasai@DESKTOP-N6MPCBS MINGW64 ~/techie-05/techie-05 (main)
$ git push -u origin main

```

 Arunasai873 Merge branch 'main' of github.com:Arunasai873/techie-05 eed7cc5 · now 🕒 5 Commits		
 .gitignore	added .gitignore	6 minutes ago
 hotfix	Create hotfix	11 minutes ago
 password	added password	12 minutes ago
 test	test	3 hours ago

Files

main
+

Go to file
t

.gitignore

hotfix

password

test

techie-05 / .gitignore

 **Arunasai873** modified

Code

Blame

2 lines (2 loc) · 16 Bytes

1 password

2 secret

13. Make a commit and make that commit reset without savings changes.

```
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app/techie-05/techie-05 (sai)
$ git log
commit db274a47abdc701c6ce6a9bf0b803d62cc96311e (HEAD -> sai)
Author: Arunsai <arunsaimailme17@gmail.com>
Date: Wed Apr 8 19:11:53 2026 +0530

    Revert "Create README.md"

    This reverts commit 728ba21c141079ac9b9ba122d1faf659ee6b4a17.

commit d2c85c507a40274f0456b1d66cbb5993950ca8f3 (origin/sai, origin/main, origin/develop)
Author: Arunsai <arunsaimailme17@gmail.com>
Date: Wed Apr 8 18:04:09 2026 +0530

    Delete tech05

commit d42708ee04912f1089c93c316377d4fcc97e73d2
Author: Arunsai <arunsaimailme17@gmail.com>
Date: Wed Apr 8 18:03:11 2026 +0530

    Create tech05

commit 144ca41bbc63eefc89d7ab7884faa46b167a2aeb
Author: Arunsai <arunsaimailme17@gmail.com>
Date: Wed Apr 8 17:40:59 2026 +0530

    Delete README.md

commit 728ba21c141079ac9b9ba122d1faf659ee6b4a17
Author: Arunsai <arunsaimailme17@gmail.com>
```

```
Revert "Create README.md"

This reverts commit 728ba21c141079ac9b9ba122d1faf659ee6b4a17.

# Please enter the commit message for your changes. Lines starting
# with '#' will be ignored, and an empty message aborts the commit.
#
# On branch sai
# Your branch is up to date with 'origin/sai'.
#
# Changes to be committed:
#   deleted:    README.md
#
~
~
~
~
~
~
~
~
~
~
```

14. Revert a committed commit to the older version

```
commit 43e025b3a47ed32143791b709c0b017bd71797db (HEAD -> master, origin/master, origin/HEAD)
merge: ba56011 38d8dbd
author: Arunsai <arunsaimailme17@gmail.com>
date: Tue Apr 7 17:09:40 2026 +0530

    Merge branch 'master' of github.com:Arunsai873/techie-horizon-05

commit ba56011d8ecb534dd1cfd8bc698cb3576d736e1f
author: Arunsai <arunsaimailme17@gmail.com>
date: Tue Apr 7 16:57:49 2026 +0530

    added as file abc

commit 38d8dbd756ef8529a030815fa577a1a41240ebfc
author: Arunsai <arunsaimailme17@gmail.com>
date: Tue Apr 7 16:56:12 2026 +0530

    file on central repo

commit 4818dfc84309e993507fb68157f9fd93e2e33cf9
author: Arunsai <arunsaimailme17@gmail.com>
date: Tue Apr 7 16:46:12 2026 +0530

    Deleted dockerfile
```

```
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/techie-horizon-05 (master)
$ git revert 4818d
[master cd7c1d0] Revert "Deleted dockerfile"
1 file changed, 8 insertions(+)
create mode 100644 dockerfile

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/techie-horizon-05 (master)
$ ls
a abc b c d dockerfile e f g h i j k m n p techie x z

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/techie-horizon-05 (master)
$
```

15. Push a file to stash without savings the changes and work on another file.

```
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ ls
alphabet hello hotfix master

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git stash
Saved working directory and index state WIP on master: 83d91b0 changed the commit message using amend command

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ ls
hello hotfix master
```



```

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git stash list
stash@{0}: WIP on master: 83d91b0 changed the commit message using ammd comman
d

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git stash pop
On branch master
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   alphabet

Dropped refs/stash@{0} (32a3106c5665a019709bb9e9af2c91d15a056588)

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ la
bash: la: command not found

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ ls
alphabet  hello  hotfix  master

```

16. Undo the stash file and start working on that again.

```

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git stash list
stash@{0}: WIP on master: 83d91b0 changed the commit message using ammd comman
d

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git stash pop
On branch master
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   alphabet

Dropped refs/stash@{0} (32a3106c5665a019709bb9e9af2c91d15a056588)

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ la
bash: la: command not found

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ ls
alphabet  hello  hotfix  master

```

```

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ ls
alphabet  hello  hotfix  master

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ cat alphabet
hello world

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git checkout release
Switched to branch 'release'

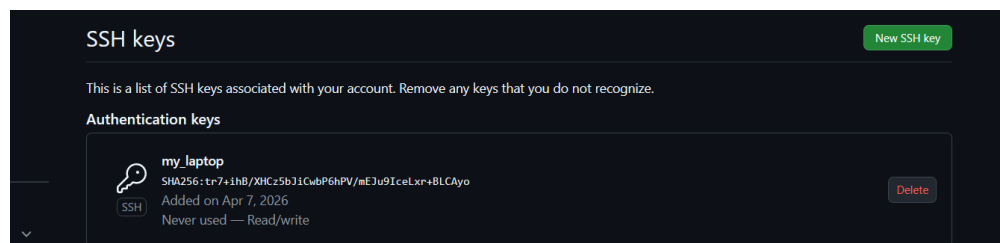
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (release)
$ ls
hello  release

```

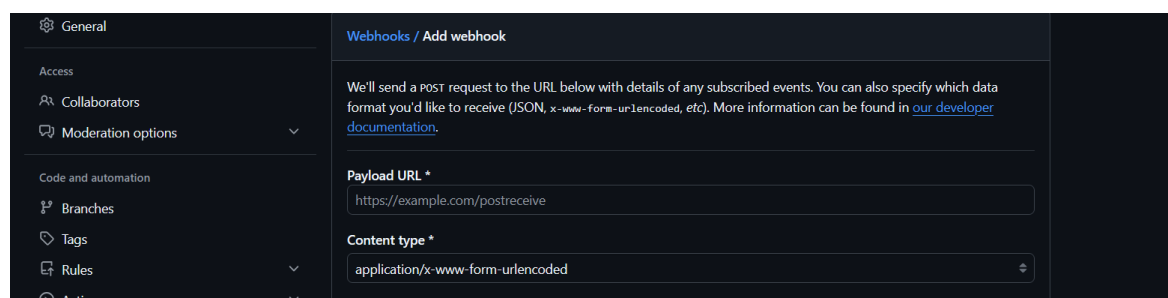
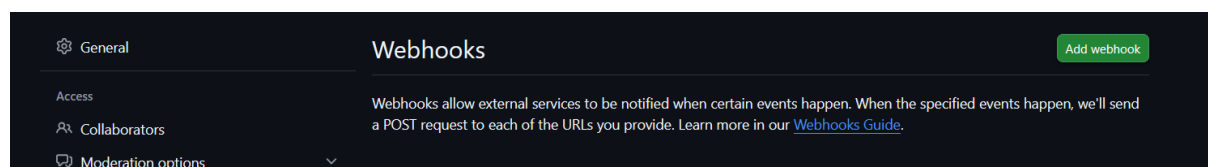
17. Generate a ssh-keygen and configure into github.

```
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ ssh-keygen
Generating public/private rsa key pair.
Enter file in which to save the key (/c/Users/LENOVO/.ssh/id_rsa):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /c/Users/LENOVO/.ssh/id_rsa
Your public key has been saved in /c/Users/LENOVO/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:tr7+ihB/XHCz5bJiCwbP6hPV/mEJu9IceLxr+BLCAyo Arunsai@DESKTOP-N6MPCBS
The key's randomart image is:
----[RSA 3072]-----
      .. O .
     . oo =
    . oo. +So+.
   E  =*oo*o+o
    . .+****.
    . ++*o.
    .o..*O=.
-----[SHA256]-----

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ cat /c/Users/LENOVO/.ssh/id_rsa.pub
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQgQDegcsEqxfZaLoSmK1Wv+LJfFNGX8HkMO2sOnTA00h2ym1tcQt3dY8nKG
7r2NIu2dbcDhq6mhtgVrTPJa9evuaUU47B1bMwDfd4ct4HPLsSJCawebh52svAKBY3zQcV/Ry0ahVdRfq+HEhSbARmHFFq9
Fy4wU4Rb7ehFDYnB3NySgrsCE+qJyod7LPVqvo06S5Zwnvj7EL2zfVnQavrNMH7qDxIjjdVrH6d5e6hrp10dI8xCVA2VV11
TRKsuwQJHRQL4EYrAwAkThLJqXLGxs6ZqgerL3Kn74AvtqALzDv1w77QZ7yTFdvbMMqfyDUy1CDBr11pS7QswRLBLTbuM+
TzhZugjgo1hZczBto1b9nsywT41czDkE9pGxCxc51vaMfFpaQCrN1xfCBChBrj91jBbZtMP1YVcIzIMnVnfPwbiwtEe1kTt
L5Kc9rbd/MkjpnUhbSjTMsMKJywyBvB9XLhLrZ05FQq+V3EkFYIH5Yaxh1yXSYSaj5/wWhyXupQU= Arunsai@DESKTOP-N
6MPCBS
```



18. Configure webhooks to github.



19. Basic understanding of .git file.

```
Arun sai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ cd .git

Arun sai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app/.git (GIT_DIR!)
$ ls
AUTO_MERGE      HEAD          config         hooks/  info/  objects/
COMMIT_EDITMSG  ORIG_HEAD    description    index   logs/  refs/

Arun sai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app/.git (GIT_DIR!)
$
```

20. Check all the logs of git.

```
Arun sai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app/.git (GIT_DIR!)
$ git log
commit d6e7854452bc137cf7d2973b0ab053f817555da7 (HEAD -> master, origin/master)
Author: Arun sai <arunsaimailme17@gmail.com>
Date:   Wed Apr 8 17:45:11 2026 +0530

    a added in master

commit 1ae2f1ddd7acdebb153a581cdc773256efe4782d
Author: Arun sai <arunsaimailme17@gmail.com>
Date:   Wed Apr 8 17:21:16 2026 +0530

    two files added

commit 7c2a1607a10af0c612f6ff25ee2b4aef26623930
```

21. Rename the commit message.

```
Arun sai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git commit --amend -m "New Message"
[master 1fca2e5] New Message
Date: Wed Apr 8 17:45:11 2026 +0530
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 a
```

```
Arun sai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app
$ git log
commit 07e764b0a4d4af485e14dcbfd5200fdf07f61376 (HEAD
Author: Arun sai <arunsaimailme17@gmail.com>
Date:   Wed Apr 8 17:45:11 2026 +0530

    Created new message
```

Created new message

```
# Please enter the commit message for your changes. Lines starting
# with '#' will be ignored, and an empty message aborts the commit.
#
# Date:      Wed Apr 8 17:45:11 2026 +0530
#
# On branch master
# Your branch and 'origin/master' have diverged,
# and have 1 and 1 different commits each, respectively.
# (use "git pull" to merge the remote branch into yours)
#
# Changes to be committed:
#   new file:   a
#
# Changes not staged for commit:
#   modified:   hello
#
# Untracked files:
#   techie-05/
```

22. Merge multiple commits into single commit.

```
Arunasai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git log -4
commit 07e764b0a4d4af485e14dcbfd5200fdf07f61376 (HEAD -> master)
Author: Arunsai <arunsaimailme17@gmail.com>
Date:   Wed Apr 8 17:45:11 2026 +0530

    Created new message

commit 1ae2f1ddd7acdebb153a581cdc773256efe4782d
Author: Arunsai <arunsaimailme17@gmail.com>
Date:   Wed Apr 8 17:21:16 2026 +0530

    two files added

commit 7c2a1607a10af0c612f6ff25ee2b4aefa6623930
Merge: 3c4bf46 da85fde
Author: Arunsai <arunsaimailme17@gmail.com>
Date:   Wed Apr 8 15:25:01 2026 +0530

    added

commit 3c4bf46cb2bfd2e9deb85fc7a7f7015e6f1d0ee9
Author: Arunsai <arunsaimailme17@gmail.com>
Date:   Wed Apr 8 15:23:08 2026 +0530

    added file techie in techie
```

```
pick c433502 added n=in release
squash f430e34 Added alphabet file
squash da85fde added file techie in release
squash 1ae2f1d two files added
squash 07e764b Created new message
squash 3fcb2f0 entered

# Rebase 3c4bf46..3fcb2f0 onto 3c4bf46 (6 commands)
#
# Commands:
# p, pick <commit>      use commit
```

```
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master|REBASE 2/6)
$ git log -4
commit f990a9262803440ed49732eef8eba88eb925b8d9 (HEAD)
Author: Arunsai <arunsaimailme17@gmail.com>
Date:   Wed Apr 8 12:51:38 2026 +0530

    added n=in release

commit 3c4bf46cb2bfd2e9deb85fc7a7f7015e6f1d0ee9
Author: Arunsai <arunsaimailme17@gmail.com>
Date:   Wed Apr 8 15:23:08 2026 +0530

    added file techie in techie

commit 0a1c500931f3076e2d005266f9fbeb01f8f83e08
Author: Arunsai <arunsaimailme17@gmail.com>
Date:   Wed Apr 8 15:05:14 2026 +0530

    alphabet

commit 5f133cdbbb0587ec5880e9689d2c302ee016394c
Author: Arunsai <arunsaimailme17@gmail.com>
Date:   Wed Apr 8 15:00:32 2026 +0530

    hello
```